UGent, 2026, pp. 1–12

Encoding Metabolic Networks with Hyperdimensional Computing

Jan P. Nawrocki, Pieterjan Vagenende, Wouter Vanhoecke and Hendrik Willems

Supervisors: Carlos Vigil-Vásquez and Michiel Stock

Abstract

In the era of big data biology, genome-scale metabolic models (GEMs) have become invaluable tools for understanding cellular metabolism across diverse organisms. These models provide a comprehensive view of metabolic networks, encompassing the complex interplay of enzymes, metabolites, and reactions. However, comparing and analyzing these intricate models in a biologically meaningful manner remains a challenging task. To address this, we introduce the application of Hyperdimensional Computing (HDC) graph encoders to classify cancer subtypes using tumor metabolic networks (TumorMet). We benchmarked native single-pass HDC classification against traditional machine learning estimators (SVM and LR) trained on hypervector feature embeddings, and evaluated retraining schemes designed to improve native accuracy. We demonstrate that HDC efficiently embeds the hierarchical structure and constituent characteristics of these biological graphs into hypervectors for downstream learning tasks. Our findings demonstrate that HDC-generated embeddings successfully classify tumor subtypes, offering a robust alternative to established baselines. While native centroid-based classification currently encounters performance plateaus due to noise accumulation within dense metabolic graphs, the representational power of HDC as a feature embedding is highly effective. Ultimately, HDC emerges as a highly promising, scalable framework for network-based oncology, highlighting a clear pathway for future optimization techniques to achieve fully transparent, "white-box" biological classification.

Keywords Genome-scale metabolic models, Hyperdimensional computing, Graph representation learning, Systems biology, Metabolic network analysis, Cancer

Abbreviations **GEMs** (Genome-scale Metabolic Models), **HDC** (HyperDimensional Computing), **HV** (HyperVector), **LR** (Logistic Regression), **ML** (Machine Learning), **PDGSMM** (Patient-Derived Genome-Scale Metabolic Models, **ssHDC** (Semantic Surprise HyperDimensional Computing), **SVM** (Support Vector Machine)

Introduction

Advancements in high-throughput "Big Data" generation and multi-omics integration have facilitated the development of genome-scale metabolic models (GEMs)—mathematical frameworks that consolidate the known metabolic information of a biological system. By providing a comprehensive scaffold of metabolic reactions, GEMs are utilized to construct networks that illuminate complex genotype–phenotype relationships (28). Given the intricate nature of metabolic dysfunction in high-complexity pathologies, network-based approaches are particularly critical.

An important application of metabolic networks is the study of cancer, a disease characterized by complex genetic and phenotypic hallmarks (15). To meet the extreme bio-energetic demands of sustained proliferation, neoplastic cells fundamentally rewire their metabolism. This dysregulation is best exemplified by the *Warburg effect*, in which cancer cells switch from efficient oxidative phosphorylation to rapid glycolysis to fuel continuous growth (15). Beyond this generalized shift, tumors exhibit profound tissue- and stage-specific metabolic plasticity (10). Mapping this heterogeneous metabolic landscape is essential for differentiating cancer subtypes and advancing personalized therapies (30).

Recently, Granata *et al.* (13) utilized GEMs to construct metabolic networks that represent the complexity of the cancer metabolic machinery. In these models, nodes represent metabolites, enzymes, or reactions, while edges denote biochemical relationships weighted by gene expression values (13).

Although these networks capture the intricacies of cancer metabolism, their structural density presents a significant challenge for analysis. Consequently, machine learning (ML) methods have become essential for interpreting this type of data (28), with dimensionality reduction and feature extraction techniques generally serving as prerequisites for a meaningful analysis. Granata *et al.* (13) addressed this by employing statistical topological descriptors and neural network-based vector embeddings (graph2vec) to facilitate cancer-type classification. These features were subsequently processed via traditional ML methods, including support vector machines (SVMs) and logistic regression (LR). While these methods demonstrate high predictive performance, they present critical limitations, such as substantial computational overhead and high sensitivity to noise in graph structures, necessitating rigorous preprocessing. Most importantly, these approaches often result in a "black box" effect; the loss of biological context during processing

Received: 13 May 2026
© The Author(s) 2026. Copyright and licence statements to be updated by the publisher during production.

obscures the interpretability required to identify specific metabolic drivers of the disease.

Due to the importance of this type of data, we believe that better alternatives should exist to analyze it. Here, we demonstrate that Hyperdimensional Computing (HDC)—an innovative approach that preserves biological interpretability while significantly reducing computational demand—could serve as a novel alternative to combat the difficulties we previously mentioned.

Hyperdimensional Computing

A comprehensive overview of HDC within the bioinformatics domain is provided by Stock *et al.* (36) and Kleyko *et al.* (23). To illustrate the fundamental mechanics of HDC, consider the conceptual representation of an apple, a pear, and a carrot (Figure 1). These objects can be decomposed into attributes: an apple is characterized by its color, category, and shape—expressed mathematically as (color $\otimes$ green) + (category $\otimes$ fruit) + (shape $\otimes$ round). Similarly, a pear is defined as (color $\otimes$ green) + (category $\otimes$ fruit) + (shape $\otimes$ oval), while a carrot is represented as (color $\otimes$ orange) + (category $\otimes$ vegetable) + (shape $\otimes$ tapered). By analyzing these overlapping traits, it is computationally evident that the apple and the pear are more similar, as they share both color and biological category.

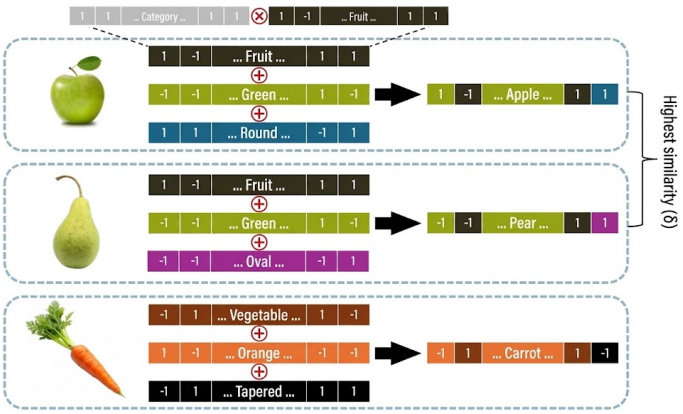


Figure 1 Illustration of the fundamental mechanics of HDC via the conceptual representation of an apple, a pear, and a carrot.

HDC implements this logic by mapping each discrete attribute to a random, high-dimensional hypervector (HV). In this paradigm, an atomic unit of information is stored as a distributed representation within a fixed high-dimensional space—typically, Dimensions (D) = 10,000 (36). Using the vast orthogonal capacity of high-dimensional space, complex computational tasks can be executed using simple vector algebra while fully preserving the semantic relationships between the original attributes. Typically, the following basic operations are utilized (36):

1. **Hypervector generation**: Each discrete attribute is mapped to a randomly initialized *atomic vector*, which may be binary, bipolar, ternary, or real-valued. In high-dimensional spaces, randomly generated vectors are almost certainly quasi-orthogonal (23), allowing a large number of unique symbols to be represented simultaneously.
2. **Bundling**: The component-wise addition of hypervectors (HVs) (+) produces a vector similar to all elements in the set. For example, bundling *green*, *fruit*, and *round* yields the composite "apple" HV.
3. **Binding**: Binding ($\otimes$) associates two HVs into a representation dissimilar to both inputs, enabling hierarchical structuring and preventing the "superposition catastrophe" where trait identities are lost within a bundle.
4. **Permutation**: A cyclic shift (ρ) introduces order during encoding, ensuring that sequences sharing identical components (e.g. ATTGC and ACGTT) yield distinct HVs.

Although these four basic operations handle symbolic and structural data, numerical information—such as the continuous gene expression values weighting the edges in the TumorMet networks—requires specialized scalar encoding techniques. For example, level encoding utilizes intervals or *bins* to discretize scalar weights into correlated HVs in order to represent these continuous data points (36).

By integrating these numerical encoding strategies with the algebraic logic of HDC, the framework transforms complex metabolic interactions into robust HVs. This holographic representation builds on a growing body of work in HDC graph encoders that have proven effective in representing discrete, continuous, sequential, and interaction data with high robustness and low computational overhead (39; 4; 6; 27; 32). Crucially, because these operations intrinsically preserve spatial similarities, the resulting model maintains a high level of structural transparency. Although outside the primary scope of this study, this transparency theoretically allows graph-level predictions to be mathematically traced back to specific node signatures, offering a distinct advantage over "black-box" machine learning methods in the characterization of cancer subtypes.

Classification

Once the graphs are encoded, they can be utilized for downstream classification. The foundation of standard HDC classification involves bundling all training HVs associated with a specific label to form a single-class HV, commonly referred to as a prototype (36). Unseen patient HVs are then classified by computing a similarity metric—typically cosine similarity—against these prototypes and assigning the label of the closest match. Although this "single-pass" training approach is computationally lightweight and highly efficient, it frequently leads to prototype saturation, a phenomenon where the bundling of background noise washes out critical discriminative patterns (17). In cancerous metabolic profiles, this is particularly problematic, as subtle variations in the pathways often determine the classification of specific subtypes within the same tissue (30).

To overcome this limitation and restore predictive performance, recent research has shifted toward dynamic frameworks commonly called "retraining schemes" (40). The basis of retraining lies in iteratively refining the class prototypes. When an HV ($\mathbf{Q}$) is misclassified, it is subtracted from the incorrect label ($\mathbf{C}_w$) and added to the correct label ($\mathbf{C}_c$), scaled by a learning rate (α) (36): $\mathbf{C}_w = \mathbf{C}_w - \alpha\mathbf{Q}$ and $\mathbf{C}_c = \mathbf{C}_c + \alpha\mathbf{Q}$. Based on this concept, several dynamic frameworks have been proposed depending on the goal (18; 17; 35; 33).

Despite its computational advantages, standard HDC faces significant challenges when applied to highly entangled datasets. Furthermore, the explicit encoding and classification of cancer-specific metabolic networks within an HDC framework remains

largely unexplored. Addressing this gap yields the primary research question of this study: *How can hyperdimensional computing be adapted to efficiently encode and classify structurally dense cancer metabolic networks?*

We hypothesize that HDC provides an effective framework for the encoding and classification of complex metabolic graphs, with its predictive performance significantly influenced by the choice of Vector Symbolic Architecture (VSA, how we encode the graphs), dimensionality, feature density, and retraining schemes. Using the results of the TumorMet study (13) as a performance baseline, we evaluated native HDC against traditional machine learning models (SVM, (5), and LR (1)) trained specifically on our generated HVs. This comparison allows us to investigate whether the primary strength of HDC lies in its native classification or in its representational power when HVs are utilized as feature embeddings. Finally, to address potential performance plateaus in highly overlapping cancer datasets, we introduce Semantic Surprise HDC (ssHDC)—a novel iterative framework designed to refine subtype differentiation and enhance signal capture.

Material and Methods

TumorMet Repository

The TumorMet repository is a comprehensive database of tumor metabolic networks derived from context-specific GEMs (13). It serves as a benchmark for graph-based ML algorithms. A primary advantage of this repository is its strict focus on patient-specific heterogeneity, which provides an ideal test to classify different cancer subtypes. The core idea is to integrate individual transcriptomic data into established metabolic scaffolds. Within this framework, our primary focus is on metabolite-based networks, where individual metabolites act as nodes and biochemical reactant–product relationships define the edges. All datasets are provided in GraphML format, a flexible, XML-based standard for storing directed, weighted graphs and associated metadata.

The datasets are organized into two distinct categories of metabolite-based networks: tissue-derived and patient-derived genome-scale metabolic models (PDGSMM). The enzyme-based and reaction-based PDGSMM networks given in the TumorMet repository are not studied in our research. Tissue-derived datasets are built from tissue-specific scaffolds and divided into seven subcategories that represent six primary tumor sites: brain, kidney, lung, ovary, prostate, and breast. These graphs share the same topological structure in terms of nodes and their connections, but differ in edge weights between patients. In contrast, PDGSMM networks are reconstructed directly from individual patient models, producing unique graph structures and weights for each sample, which presents a significantly more complex classification challenge.

Data Ingestion and Preprocessing

We implemented HDC-encoding pipelines in Python (37) using the Torchhd library (v5.8.4 (16)). TumorMet's network GraphML files were loaded and their associated edge attributes managed using the NetworkX library (v3.4.2 (14)). For the representation-based classification procedures, we utilized the Scikit-learn library (v1.6.1 (29)). To facilitate supervised classification, the clinical metadata provided by the TumorMet repository (13) was mapped using unique sample identifiers.

Table 1 Metabolites-based Network Data for Selected Tissues. The Count column gives the number of graphs for the tissue-derived and the PDGSMMs-derived networks respectively.

Tissue	Graph Network Type	Count	Cancer subtypes
Ovary	Tissue-derived	290	Proliferative, mesenchymal,
	PDGSMM	230	differentiated, immunoreactive
Prostate	Tissue-derived	497	Gleason Pattern 3, 4, 5
	PDGSMM	422	
Lung	Tissue-derived	1025	Adenocarcinoma (LUAD),
	PDGSMM	829	Squamous cell carcinoma (LUSC)
Brain	Tissue-derived	666	Glioblastoma multiforme (GBM),
	PDGSMM-derived	138	Lower grade glioma (LGG)
Kidney	Tissue-derived	822	Kidney Renal Papillary (KIRP),
	PDGSMM-derived	737	Kidney Renal Clear cell (KIRC)
Breast	Tissue-derived	1085	Basal-like, HER2-enriched,
	PDGSMM-derived	910	Luminal A, Luminal B, Normal-like

To ensure the biological relevance of the classification task and mitigate potential batch effects, "Primary Tumors" were selected exclusively for all analyses. Finally, we labeled the samples according to Table 1, and the samples were divided into subtypes according to the diagnostic categories defined by Granata *et al.*

For the tissue-derived graphs, all classification tasks have been performed except for breast tissue, because the subtypes are not defined in the TumorMet's sample sheet (additional preprocessing is needed that goes beyond the scope of this project). For the PDGSMM-based graphs, the greatest attention has been paid to the kidney tissue, as it was the only tissue for which classification scores are given in the original study. To evaluate our approach, we compared our predictive performance with the baselines provided by TumorMet, which was established using the SVM classification based on Jensen-Shannon graph distances (9) and graph2vec encoding (26) for the tissue-derived and PDGSMM graphs, respectively.

Encoding

To encode the graphs into HVs, we utilize the Multiply-Add-Permute (MAP) Vector Symbolic Architecture (VSA). In this implementation, the initial random base HVs are generated by sampling from a bipolar distribution, where each element is chosen from {-1, 1} with equal probability. This ensures that the initial vectors are approximately orthogonal in high-dimensional space. As the encoding process involves the accumulation of these vectors through the "Add" (superposition) operation, the resulting elements in the HV can range from (-infinity, +infinity). To maintain stability and facilitate downstream tasks, these values are typically normalized to a range of -1 to 1 before the final classification.

Table 2 Comparison of Encoding Strategies for Tissue-derived vs. PDGSMM-derived Networks.

Feature	Tissue-derived	PDGSMM-derived
Graph types	Weighted directed	Unweighted undirected
Topology	Homogeneous	Heterogeneous
Variable	Edge weights	Topology
Encoding Focus	Global	Neighborhood

We will classify two categories of networks by encoding these networks into HVs. Table 2 describes the differences between the categories and how to encode them.

Weighted Encoding: Tissue-derived Graphs

For the tissue-derived graphs, our encoders were fundamentally based on GrapHD (32), as it is the only graph encoder that can encode weighted graphs.

We implemented four architectural modifications to GrapHD to handle the extreme density of the TumorMet graphs:

- We replaced GrapHD's HV-flipping mechanism with a level encoder for edge weights. This approach is highly computationally efficient, allowing direct mapping to precomputed level HVs rather than requiring iterative bit-flipping of weight representations.
- Instead of constructing intermediate neighborhood memories, we encoded edges directly through the binding of source, target, and weight HVs. While mathematically equivalent to the standard GrapHD approach due to the distributive property of HDC operations, this method scales more effectively for dense networks by eliminating the redundant re-computation of node interactions across overlapping neighborhoods.
- To preserve directionality, we applied permutation to individual nodes during edge-binding rather than permuting bundled neighborhoods. This was required by our modified architecture, which bypasses the need to construct intermediate neighborhood structures.
- To prevent HV saturation caused by thousands of edges, we prioritized noise filtration *before* bundling, effectively eliminating the need for GrapHD's post-encoding memory refinement.

We evaluated two different encoder variants. Both follow a three-stage pipeline—Filtering, Normalization, and Vector Composition—but differ in how they prioritize edge importance and normalize weights.

1. **Simple Encoder**: Utilizes absolute thresholding and linear quantization. It prioritizes high-magnitude connections by filtering edges based on dataset-wide quartiles. The weights are normalized via Min-Max scaling to a linear range of $[0, 1]$.
2. **Statistical Encoder**: Uses relative thresholding based on variance-based feature selection. It assumes that the most informative edges are those that exhibit the highest variability between training samples. The weights undergo Z-score normalization and a sigmoid transformation to map the distributional deviations into the $[0, 1]$ range.

The Simple Encoder is optimal for identifying the heaviest metabolic reactions, while the Statistical Encoder excels at isolating the most informative reactions (those that differ the most significantly between patients). Algorithm 1 generalizes this process. The specific implementation of the functions `Filter` and `Normalize` depends on the variant chosen.

To support the use of variance as a biologically meaningful selection criterion, we conducted a feature interpretation analysis on the lung dataset. Specifically, the top 1% of the edges ranked by variance were extracted and mapped to their pairs of constituent metabolites.

Furthermore, to evaluate computational efficiency, we conducted a comparative timing analysis between the Simple Encoder—utilizing various dimensionalities and median edge-weight filtering—and a traditional SVM baseline. For the baseline model, graph data were vectorized by computing the row-wise sums of the adjacency matrices to generate node-strength profiles, which were then normalized by standardization of the Z-score and classified using a linear-kernel SVM. To ensure strict timing accuracy and mitigate hardware noise, both methods were executed on identical hardware, with all processing times averaged over 10 independent runs using fixed random seeds.

Algorithm 1 GrapHD-variants encoding

Require: Graph $G = (V, E)$, Weights W, Threshold Type T, Node HVs $\{\mathbf{HV}_{node}\}$, Level HVs $\{\mathbf{HV}_{level}\}$
Ensure: Graph Hypervector $\mathbf{H}_G$

$\mathbf{H}_{edges} \leftarrow \emptyset$
$E_{active} \leftarrow \text{Filter}(E, T)$ ▷ Absolute Quartile vs. Relative Variance
for each edge $e = (u, v)$ with weight w in E_{active} **do**
  $w_{norm} \leftarrow \text{Normalize}(w)$ ▷ Min-Max vs. Sigmoid(Z-score)
  $bin \leftarrow \lfloor w_{norm} \cdot (\text{num_levels} - 1) \rfloor$
  $\mathbf{H}_w \leftarrow \mathbf{HV}_{level}[bin]$
  $\mathbf{H}_{edge} \leftarrow \mathbf{HV}_{node}[u] \otimes (\rho(\mathbf{HV}_{node}[v]) \otimes \mathbf{H}_w)$ ▷ Permutation of Target Node
  $\mathbf{H}_{edges} \leftarrow \mathbf{H}_{edges} \cup \{\mathbf{H}_{edge}\}$
end for
$\mathbf{H}_G \leftarrow \text{normalize}(\sum \mathbf{H}_{edges})$ ▷ Bundling

Unweighted Encoding: PDGSMM Graphs

For the PDGSMM-derived graphs, we used only unweighted encoders, as the original TumorMet article reported decreased performance with weighted variants on these specific kidney networks (13). The selected encoding frameworks included GrapHD (32), GraphHD (27), and the star subgraphs encoder proposed by Vergés *et al.* (39).

While GraphHD is similar to GrapHD—binding node HVs to represent edges before bundling them into a global graph HV—it gives nodes that have the same PageRank (a popular centrality metric in network science) the same initial node representation. Given the fact that it is rare that two nodes have the same PageRank, PageRank values were rounded to a fixed number of decimal places to facilitate grouping. We perform an analysis of this rounding parameter to evaluate its impact on classification accuracy.

In contrast, the star subgraphs encoder (Algorithm 2) represents the local topology by capturing the immediate neighborhood surrounding each node. During encoding, each central node representation is bound with the representations of its adjacent neighbors. These localized "star" HVs are subsequently bundled to produce the final graph-level representation.

Algorithm 2 Star subgraph encoding

Require: Training set of N labeled graphs $\{G_i\}_{i=1}^{N}$
Ensure: Class prototype hypervectors $\mathbf{HV}_{\text{class}}$

for $i = 1 \ldots N$ **do**
  $G \leftarrow G_i$
  $\mathbf{HV}_G \leftarrow \mathbf{0}$
  for each node v in G **do**
    $\mathbf{HV}_{temp} \leftarrow \text{encoding}(v)$
    for each neighbor $u \in \text{neighbours}(v)$ **do**
      $\mathbf{HV}_{temp} \leftarrow \mathbf{HV}_{temp} \otimes \text{encoding}(u)$ ▷ Binding
    end for
    $\mathbf{HV}_G \leftarrow \mathbf{HV}_G + \mathbf{HV}_{temp}$ ▷ Bundling
  end for
  $\mathbf{HV}_{\text{class}} \leftarrow \text{adaptive_learning_algorithm}(\mathbf{HV}_G, i, \mathbf{HV}_{\text{class}})$
end for
return $\mathbf{HV}_{\text{class}}$

Classification

After encoding the graph data into high-dimensional space, we evaluated a variety of classification strategies on the resulting HVs. Here, we tested a native HDC associative memory approach (single-pass), two traditional machine learning classifiers (LR, logistic regression, and SVM, support vector machine), and dynamic HDC retraining schemes. These classification paradigms were implemented for both problem types (tissue-derived and PDGSMM graphs).

All models were evaluated using a standardized training, optimization, and testing pipeline. For the tissue-based classification tasks, we decided to evaluate the classification models using the Macro-F1 score because it combines precision and recall into a single metric and is inherently capable of handling class imbalance. For the PDGSMMs-based kidney classification task, the weighted F1 score was used. For the tissue-derived graphs, optimal hyperparameters were identified using a 10-fold stratified cross-validation grid search (Table 3). For the PDGSMM graphs, 5-fold cross-validation was used during grid search to reduce computational load. This is because the encoders used for the PDGSMM graphs take longer to compute compared to the tissue-derived versions. Once validated, the combination of the best-performing parameters was used to encode the data, train the final model on the entire training set, and evaluate its performance on an isolated test set.

Table 3 Hyperparameter grid evaluated for each encoding strategy.

Encoder	Parameter	Tested Values
Simple Encoder	Dimensions (D)	1000, 5000, 10000
	Quartile Threshold	Q1, Median, Q3
Statistical Encoder	Dimensions (D)	1000, 5000, 7500, 10000
	Variance Threshold	10%, 50%, 90%
Star Subgraphs	Learning Rate	10, 20, 30, 40, 50
	Learning Rate Decay	0.75, 0.8, 0.85, 0.9, 0.95
ssHDC	Variance Threshold	5%, 90%
	Learning Rate (α)	0.05, 0.1, 0.2, 0.5, 1
	Margin (THR)	0, 0.005, 0.1

Single-pass Centroid Classification

The single-pass approach serves as the standard classification paradigm in HDC. During the training phase, a "prototype" or "class HV" is generated for each label by performing an element-wise summation of all encoded training graphs belonging to that class. While the Simple Encoder normalizes these resulting prototypes, the Statistical Encoder does not. For inference, a query graph is classified by calculating the cosine similarity between its encoded HV and each class prototype; the class delivering the highest similarity score is assigned as the prediction.

Traditional Machine Learning Classification

To benchmark the discriminative power of the graph HVs, LR and SVM were employed as downstream classifiers using the HV as feature vectors:

- **HDC + LR:** A linear model trained to estimate the probability of class membership using HV elements as input features.
- **HDC + SVM:** A linear-kernel SVM utilized to identify the optimal hyperplane that maximizes the margin between the high-dimensional graph representations of different classes.

By training LR and SVM models on the resulting HVs, we evaluated the linear separability of the graphs in the high-dimensional space and determined whether the HDC projection inherently enhances predictive performance compared to the native single-pass centroid approach.

Standard Retrainers

The star subgraphs encoder integrated three established iterative retraining algorithms: AdaptHD (18), OnlineHD (17), and RefineHD (38). We systematically investigated the effects of tuning the learning rate and learning rate decay for AdaptHD and OnlineHD, as these hyperparameters could influence the performance of the retraining methods.

Novel Retrainer: Semantic Surprise HDC

Extending the statistical encoder ($D = 5000$), we developed ssHDC, a neuro-inspired iterative framework designed specifically for complex, tissue-dependent datasets (e.g., ovarian cancer). ssHDC synthesizes proactive conflict monitoring (2; 7; 35) with the novelty-driven updates characteristic of OnlineHD (17). Retraining is triggered not only upon misclassification, but also when the model correctly identifies a label yet detects *pre-response conflict*—defined as the similarity margin between the correct class (δ_c) and the runner-up (δ_r) falling below a predefined threshold (THR):

$$\delta_c - \delta_r < \text{THR}. \tag{1}$$

Once triggered, the model executes an iterative update scaled by a "Semantic Surprise" factor. Drawing on cognitive models of semantic violation (24), this dynamic weight ensures that patient samples that deviate significantly from the established prototype, presenting critical or novel metabolic dysregulations, receive proportionally larger updates. The surprise factor is calculated as the distance (cos) between the patient HV ($\mathbf{Q}$) and the correct class prototype ($\mathbf{C}_{\text{correct}}$):

$$\text{Surprise} = 1.0 - \cos(\mathbf{Q}, \mathbf{C}_{\text{correct}})\,. \tag{2}$$

To execute this reciprocal update, the Surprise factor is applied along with an epoch-decaying learning rate (lr), stabilizing convergence over time:

- **Excitation** (pulling the correct prototype toward the patient vector):

$$\mathbf{C}_{\text{correct}} = \mathbf{C}_{\text{correct}} + (\mathbf{Q} \cdot \text{lr} \cdot \text{Surprise}) \tag{3}$$

- **Inhibition** (pushing the incorrectly predicted prototype away to reduce overlap):

$$\mathbf{C}_{\text{wrong}} = \mathbf{C}_{\text{wrong}} - (\mathbf{Q} \cdot \text{lr} \cdot \text{Surprise}) \tag{4}$$

Because these updates are highly fractional, immediate normalization can precipitate floating-point underflow on standard CPUs. To maintain numerical stability, ssHDC utilizes an epoch-level accumulation approach, strictly deferring prototype normalization until the end of the training loop.

Results and Discussion

Performance Benchmarking on Tissue-Derived Networks

Parameter Sensitivity

The performance of HDC highly depends on the architectural properties of its HVs. In this paper, we analyze the sensitivity of classification performance to two critical hyperparameters: HV dimensionality and edge-weight thresholding criteria. Here, we present the analysis of the ovarian tissue dataset, which was identified as the most challenging for multi-class subtype classification. All the F_1-scores, per encoder (Simple and Statistical), model (HDC, HDC +LR and HDC + SVM), D (1K, 5K and 10K), and threshold (Q1/10%, Median/50%, and Q3/90%) are given in the Supplementary Table 6. In Figure 2, this is displayed as a dot plot.

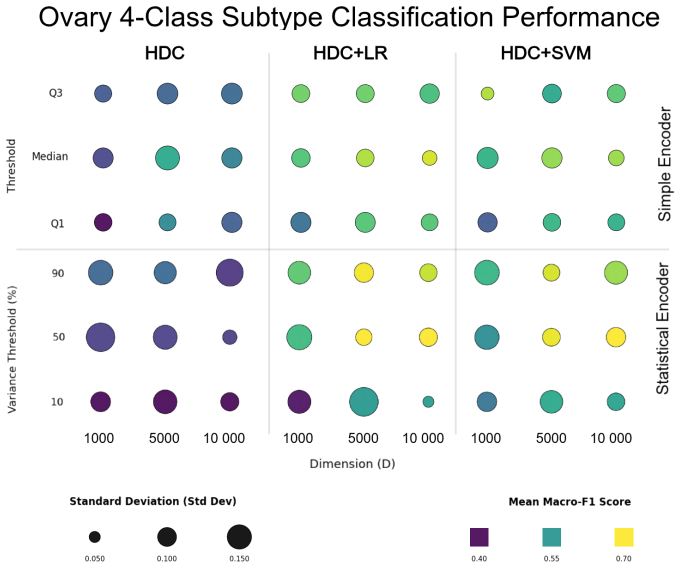


Figure 2 Ovary 4-class subtype classification utilizing the Statistical and Simple Encoders. Each panel displays different classification pipelines (HDC, HDC + SVM, and HDC + LR) alongside varying dimensionalities and edge-weight filtering parameters. The color intensity represents the mean F_1-score across 10-fold cross-validation, while the diameter of each point corresponds to the standard deviation.

The HDC-LR and HDC-SVM models are comparable in performance but consistently outperform the baseline single-pass HDC centroid classifier when evaluated per combination for both encoders. Dimensionality emerged as a primary performance driver, with higher D generally yielding higher F_1-scores. Unlike in (41), where lower D yielded higher accuracy, we see that higher dimensions lead to higher performance. This can be because in (41), they performed an easy classification task in comparison to this project, where the type and amount of data are complex and multi-class. Consequently, the grid search for the Statistical Encoder was expanded to evaluate dimensionalities between 5,000 and 10,000: since D = 1,000 failed to yield peak performance for any model configuration and performance had not yet plateaued at the bounds of the original search range, D = 7,500 was introduced as an additional data point to more finely resolve the performance curve. This value was subsequently added to Table 7 and, for several dataset cases, proved to be the optimal dimensionality.

Regarding feature selection, a Q1/10% threshold—whether based on absolute magnitude or relative variance—generally produced the less optimal outcomes for the ovary dataset. For the Simple Encoder, the median threshold generally gives the best performance, while for the Statistical Encoder, a threshold of 90% generally gives the best performance. The impact of these choices is evident when comparing specific configurations: for the Simple Encoder HDC model at D = 10,000 and the Q1 threshold, there is an F1 of 48%, while for the HDC + SVM model at D = 1,000 and the Q3 threshold, the F1 is 66%. This indicates that while the model choice is very important, the dimension and thresholding are not negligible. Ultimately, the absence of universally optimal parameter combinations across all datasets demonstrates that structural configurations are intrinsically linked to dataset metadata—such as sample size and biological complexity. Consequently, rigorous parameter tuning remains essential for robust classification.

Comparative Analysis Against the TumorMet Baseline

After parameter optimization, the best parameter (D and threshold) per model for both encoders is chosen to try to predict a final test set. In Table 7, this is displayed for every tissue with various performance metrics.

As the table demonstrates, optimal dimensionality and thresholding parameters highly depend on tissue type. The Simple Encoder now achieves optimal performance across all thresholds (in contrast to optimal performance at median thresholding for the ovary tissue). The Statistical Encoder consistently favors a 90% threshold (aligning with the ovary results). Furthermore, hybrid models (HDC + LR or HDC + SVM) continue to outperform centroid-based HDC. Generally, a dimensionality of 10K consistently yields the best performance.

Figure 3 presents a comparative analysis of the F_1-scores achieved by the optimal classification models of both the Simple and Statistical Encoders for every tissue against the benchmark values reported in the original TumorMet study (13).

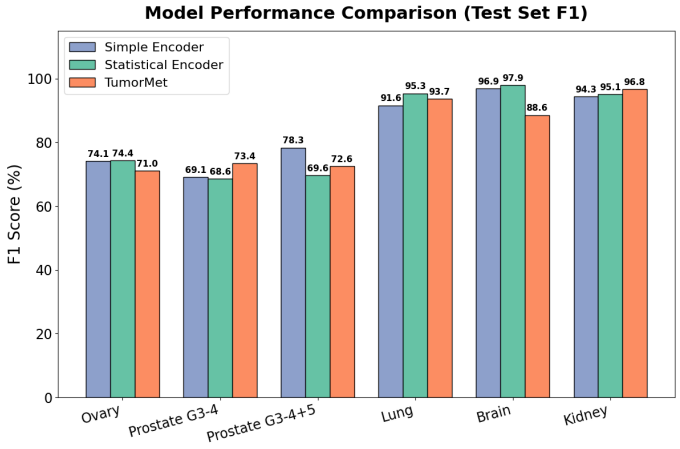


Figure 3 Performance comparison across datasets (x-axis), evaluating test set F_1-scores (y-axis) for the Simple and Statistical Encoders versus the TumorMet baseline.

Overall, the Simple and Statistical Encoders yield comparable performance, although their relative efficacy varies strictly by tissue type. The Statistical Encoder demonstrates superior accuracy in ovary, lung, and brain tissues, whereas the Simple Encoder performs better on prostate (G3–4+5) samples. Generally, one

of our encoders always outperforms the TumorMet baseline, with the notable exception of the prostate G3–4 and kidney datasets. While our approach yielded some improvements, instances where predictive performance did not surpass the baseline offer critical insights into the limitations of the current framework. While certain highly intertwined cancer subtypes (e.g., prostate G3–4) may offer minimal opportunity to improve upon low baseline F1-scores, our observations suggest a departure from metabolic profiles reported in previous studies. For instance, whereas (12) suggests that Gleason patterns 3 and 4 are metabolically distinct, our results—aligned with the TumorMet benchmark—did not reflect this differentiation. Similarly, highly distinct subtypes (e.g., kidney) appeared to reach a performance ceiling in our tests, contrasting with the findings of (3). These discrepancies may stem from specific characteristics of the dataset used or the inherent complexity of translating metabolic signatures across different experimental cohorts, rather than a definitive lack of biological variance. Our findings suggest a fundamental distinction in how different cancer subtypes organize their metabolic networks (13; 22). This distinction explains the observed performance variance: The Simple Encoder appears most effective when classification is driven by high-flux reactions captured via absolute thresholding, such as in the prostate G3–4+5 dataset. In this specific subtype, absolute flux magnitudes are likely more discriminative than relative variance; here, the Statistical Encoder may prioritize noisy variable edges, whereas the Simple Encoder captures the stable, high-flux reactions that genuinely define the pathology. Conversely, the Statistical Encoder excels in tissues where the most discriminative features are not the highest-magnitude reactions, but rather those exhibiting the highest inter-patient variance, requiring a relative thresholding strategy.

To further enhance these results, coupling HDC with traditional machine learning methods (SVM and LR) successfully improved the TumorMet baseline, as also shown in (34) where they use HV as input for feed-forward neural networks. This reliance, however, presents a structural compromise. Establishing a functional, standalone centroid model remains highly desirable because it theoretically enables direct feature interpretation. The transparency maintained by the centroid-based method allows graph-level predictions to be mathematically traced back to specific node signatures via similarity scores between individual nodes and class prototypes.

While we did not evaluate the inherent interpretability of HDC itself, we successfully established a robust biological baseline by interpreting the edge-variability preprocessing of the Statistical Encoder. The top features among the highest-variance edges were dominated by cytosol–blood transport metabolic activity, particularly amino acid transport. The highest-variance edge involved sodium ion (Na^+) transport, likely reflecting electrogenic co-transport activity coupled to amino acid uptake (Table 4). The pattern observed in the top-edge features is biologically consistent with known lung cancer metabolic reprogramming: amino acid uptake is upregulated in tumors to support nucleotide synthesis, redox balance (GSH), and the mTOR signaling pathway (25). These results confirm that the Statistical Encoder's variance-based selection captures metabolically relevant, cancer-associated transport activity rather than arbitrary noise.

To evaluate practical viability, we compared the processing time and classification performance of the HDC framework against a traditional baseline of SVM in tissue-derived graphs with varying dimensionalities and median edge-weight filtering (Figure 4). Although the absolute execution time of the SVM's remains lower than that of the HDC across the observed sample range, the empirical growth rate of the SVM of 2.37 substantially exceeds those of the HDC variants (1.32, 1.16 and 1.13 for D = 1,000; 5,000 and 10,000; respectively), indicating a more favorable computational trajectory for HDC as graphical data scales. Although HDC exhibits higher latency than standard SVMs in this implementation, it remains significantly more efficient than most Graph Neural Networks (8), and lower-dimensional HVs (e.g., D = 1,000) offer a considerable speed advantage, underscoring HDC's inherent flexibility in balancing predictive performance and computational cost.

Table 4 Top Features from the Lung Dataset Sorted by Highest Edge Variance (Statistical Encoder).

Source	Source Compound	Target	Target Compound	Variance
Na^+	Systemic (Blood)	Na^+	Cytosol	21737.71
HCO_3^-	Cytosol	HCO_3^-	Systemic (Blood)	2336.70
tryptophan	Systemic (Blood)	tryptophan	Cytosol	743.88
GSH	Cytosol	HCO_3^-	Systemic (Blood)	710.82
alanine	Cytosol	alanine	Systemic (Blood)	614.33
serine	Systemic (Blood)	serine	Cytosol	597.23
asparagine	Cytosol	asparagine	Systemic (Blood)	554.55
methionine	Systemic (Blood)	methionine	Cytosol	538.84
valine	Systemic (Blood)	valine	Cytosol	527.04
tyrosine	Systemic (Blood)	tyrosine	Cytosol	460.42

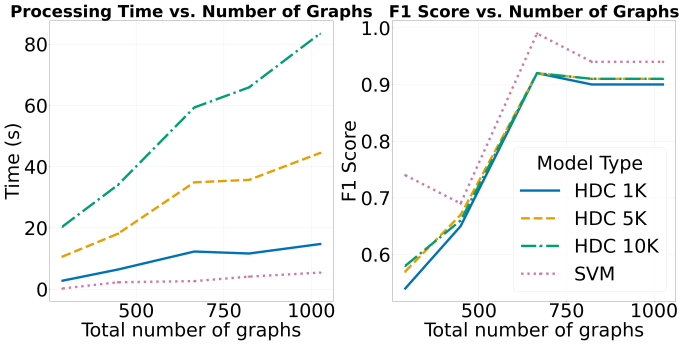


Figure 4 Processing Time and F1-score vs. Total Number of Graphs for HDC ($D = 1\text{K}, 5\text{K}, 10\text{K}$) and SVM.

Performance Benchmarking on PDGSMM-Derived Networks

Figure 5 shows the mean F_1-scores and their standard deviation obtained using the star subgraph encoder paired with either the AdaptHD or OnlineHD retraining frameworks per combination of learning rate and learning rate decay. Across both AdaptHD and OnlineHD, changing the initial learning rate has a considerably greater impact on test performance than adjusting the learning rate decay. Setting the learning rate decay parameter to 0.9 seems to give the best results and also gives the lowest standard deviation.

Using the optimal hyperparameter combinations found during this grid search, a comparison of the different encoders is made for the kidney tissue. The resulting F_1-scores are presented in Figure 6, with the TumorMet reference F_1 score (83.6%) given by a dashed red line. Similar methods are given the same color. The most basic GrapHD method achieves an F_1-score comparable to the reference baseline,

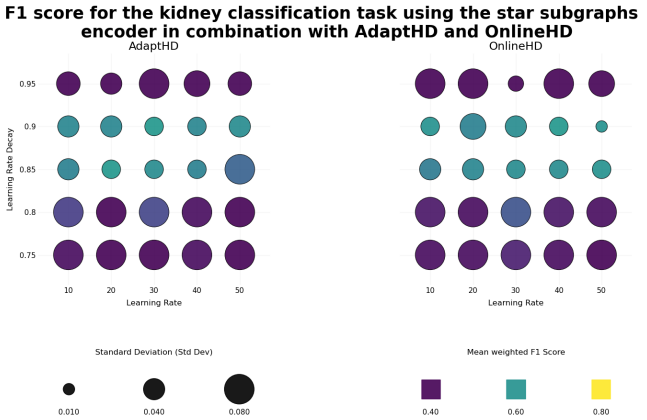


Figure 5 F_1-scores for the kidney 2-class subtype classification based on PDGSMM-derived graphs, utilizing the star subgraph encoder across varying initial learning rates and learning rate decay parameters for the AdaptHD and OnlineHD frameworks.

whereas augmenting GrapHD with SVM or LR yields scores (91.0% and 89.7%, respectively) that surpass the baseline. This suggests that using HDC-based methods in combination with more classical ML approaches could be an interesting avenue to further explore.

Among the star subgraphs methods (depicted in blue in Figure 6), the F_1-score for the star subgraphs encoder without retraining (63.4%) closely aligns with those achieved using AdaptHD or OnlineHD (both 61.0%), while the RefineHD algorithm demonstrated the lowest comparative performance (17.4%). This contrasts with the findings of the publication of the star subgraphs method (39). This difference might be explained by the different nature of the datasets used.

In general, it was noted that the computations for the PDGSMM-derived graphs took remarkably longer than those for the tissue-derived.

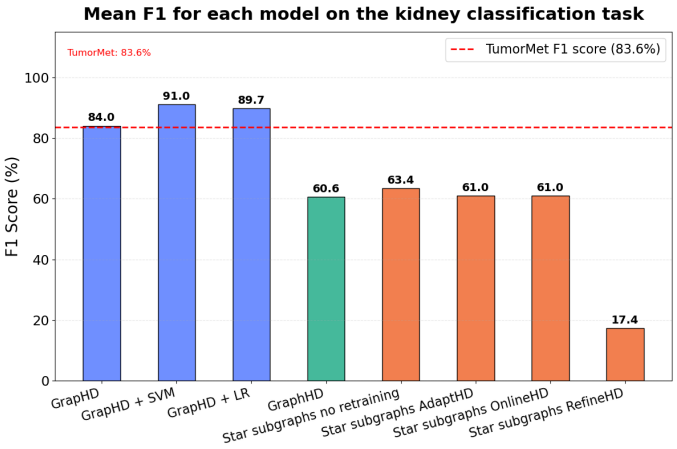


Figure 6 F_1-scores (y-axis) for the kidney 2-class subtype classification based on PDGSMM-derived kidney graphs. For each method (x-axis), the mean F_1-score following 5-fold cross-validation is reported using the optimal hyperparameter combination. The reference F_1-score from the TumorMet study is depicted by a dashed red line.

Prototype normalization effect

A critical divergence in our single-pass baselines involved prototype normalization: The Simple Encoder included prototype normalization, while the Statistical Encoder did not. This distinction proved crucial. Although the evaluated metabolic graphs share identical topologies, their edge weights (metabolic fluxes) vary substantially. Without normalization, high-flux "outlier" patients disproportionately dominate the aggregated class HVs, distorting the representative subtype signal.

An ablation study on the Ovary dataset quantified this impact. Removing normalization from the Simple Encoder degraded the optimal F_1-score identified during the parameter search from 62.05% to 54.12% (Table 5). Similarly, integrating normalization into the Statistical Encoder—evaluated at fixed parameters (D = 5000, density = 90%) as a prerequisite to support our iterative ssHDC framework—boosted its initial performance of 49.00% to 61.73% (Table **??** and Figure 7). Correcting this imbalance effectively mitigates outlier influence, balances precision and recall, and ensures robust scaling. Although the encoders utilized for the PDGSMM-derived graphs were not explicitly analyzed for this effect, these findings strongly suggest that normalization is not only an artifact of the Simple Encoder but a fundamental prerequisite for capturing accurate subtype signatures in flux-weighted HDC architectures.

Table 5 HDC Performance: Normalized vs. Non-Normalized Prototypes.

Norm.	D	Thresh.	Acc.	F1	Prec.	Rec.
Yes	5000	Median	0.632	0.621	0.621	0.626
No	1000	Q3	0.561	0.541	0.640	0.581

ssHDC

Because these optimized baseline centroid methods still fell short of the theoretical predictive ideal, we evaluated the ssHDC framework across varying edge densities and hyperparameter configurations. This iterative retraining framework is designed to enhance signal capture and refine differentiation by proactively correcting misclassification using a strict margin threshold and a "Surprise" factor weighting (17; 35).

During cross-validation, restricting the network to the top 5% of highly variable edges yielded substantial F_1-score improvements (from $38.38\% \pm 9.83\%$ to $54.83\% \pm 6.20\%$ with $\alpha = 0.2, \text{THR} = 0.0$). Conversely, retaining 90% of edges produced only marginal gains (from $61.73\% \pm 6.32\%$ to $64.50\% \pm 5.52\%$ with $\alpha = 1.0, \text{THR} = 0.0$) (Figure 7). Final evaluations on an isolated test set reinforced this dynamic: at 90% density ($\alpha = 1.0, \text{THR} = 0.0$), retraining actively degraded the test F_1-score from 64.67% to 59.20%; however, at 5% density ($\alpha = 0.2, \text{THR} = 0.0$), the framework generalized effectively to unseen data, delivering a measurable improvement (Baseline F_1: 43.70% → Retrained Test F_1: 45.43%).

Beyond edge density, individual ssHDC hyperparameters exhibited distinct optimization trends. Elevated margin thresholds (e.g., THR = 0.1) consistently degraded performance. In contrast, increasing the learning rate linearly improved predictive performance until a plateau is reached near $\alpha = 0.2$. Furthermore, performance remained remarkably robust even at maximum initial learning rates ($\alpha = 1.0, \text{THR} = 0.0$).

The contrast between these density thresholds highlights the model's vulnerability to background noise. In densely connected

graphs (90% retention), the retrainer yielded almost no improvement and even exhibited overfitting due to the persistence of uninformative edges. Rigorous filtration (5% retention) effectively mitigates this, allowing the retrainer to successfully generalize. Furthermore, hyperparameter optimization indicates that class prototypes are inherently too tightly clustered in the high-dimensional space to accommodate wide-margin proactive updates, necessitating a highly restrictive or completely disabled margin (THR $\leq$ 0.005). Nevertheless, the Semantic Surprise factor ensures that early updates remain meaningful and robust.

Ultimately, the evaluation of the ssHDC framework validates the rationale behind this design: iterative HDC retraining is effective only when the hyperdimensional space is optimized to prevent noise memorization. Hence, implementing spatial optimization frameworks—such as Graph Memory Refinement of GrapHD (32) or the selective bundling of RefineHD (38)—before ssHDC retraining could further isolate critical metabolic signals. Further mitigating baseline noise, hybrid approaches may generate refined prototypes capable of even greater precision in differentiating highly overlapping cancer subtypes.

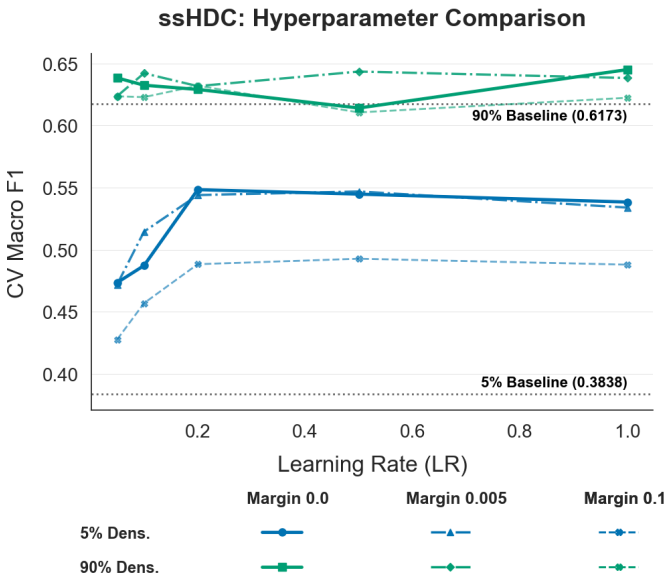


Figure 7 ssHDC Variable Edge Density Comparison: Hyperparameter sensitivity at 5% (*blue*) and 90% (*green*) edge densities, illustrating the impact of the learning rate (α) and proactive margin (THR) on the macro F_1-score.

Future work

Although this study investigates a diverse range of architectural configurations and hyperparameters, it is framed as an exploratory pilot study rather than as an exhaustive combinatorial analysis. Given the encouraging results, future work should scale this framework to a broader parameter space. Specifically, several individual variables merit isolated investigation, such as the decoupled effects of weight normalization (Min-Max vs. Z-score) versus thresholding strategies (absolute vs. relative). Furthermore, our current implementation uses a specific HV initialization; the impact of varying the underlying atomic vector types—specifically, binary, bipolar, ternary, and real-valued HVs—remains an open question. Extending these methods to additional datasets and cross-testing encoders derived from PDGSMM on tissue-derived graphs will be essential to fully characterize the robustness of the HDC approach. Ultimately, the most promising frontier lies in the transition from classification to discovery: leveraging the inherent transparency of HDC to pinpoint the specific metabolites, enzymes, and reactions that drive tumor-specific metabolic reprogramming. For future studies, more types of encoders should be used to make a complete systematic comparison as described in (21), (42), and (43). Also, different VSA architectures should be studied, like described here: (11), (19), and (31). In this study, only one VSA architecture was studied: MAP, which may not be ideal for the type and amount of data these networks have. Regarding time efficiency, future work should investigate scalability across larger datasets, with particular promise in hardware acceleration via GPUs, FPGAs, or specialized designs (20), as well as Binary Spatter Codes (19) to further reduce execution times through efficient bit-wise computation.

Conclusion

In this exploratory study, we present the first application of HDC to classify cancer subtypes using tumor metabolic networks. Our primary objective was to adapt the HDC framework to efficiently encode and classify these structurally and topologically complex biological graphs. Successfully encoding these complex biological graphs into HVs revealed a critical interplay between the chosen VSA, dimensionality, feature density, and retraining schemes. Because expanding the hyperdimensional space directly increases computational cost, finding the optimal balance between these three factors is essential to maximize predictive accuracy without sacrificing HDC's inherent efficiency. Although our findings establish HDC as a robust, scalable alternative to the baseline models utilized by Granata *et al.* (13), native centroid-based classification often struggles to match the predictive performance of traditional methods. Consequently, HDC's current primary strength lies in its representational power when HVs are utilized as feature embeddings. However, relying on external estimators sacrifices the "white-box" transparency inherent to native centroid logic. In an effort to preserve this transparency, we implemented iterative retraining schemes. However, these did not yield improvements—a plateau we hypothesize is primarily driven by noise accumulation within the dense metabolic graphs. Future work must therefore prioritize spatial optimization techniques to mitigate background noise before or during retraining. Ultimately, because each cancer dataset possesses a unique underlying biological narrative, evaluating these networks holistically is paramount. Our findings highlight that merging biological insight with the symbolic, distributed representation of HDC remains a challenging yet highly promising pathway toward harnessing the full potential of network-based cancer classification.

Data availability

The data underlying this article are available in our Github (37) at https://github.com/Kermit-UGent/DesignProject25-MetabolismHDC.

References

1. Joseph Berkson. Application of the logistic function to bio-assay. *Journal of the American Statistical Association*, 39:357–365, 9 1944.

2. Matthew M. Botvinick, Jonathan D. Cohen, and Cameron S. Carter. Conflict monitoring and anterior cingulate cortex: An update. *Trends in Cognitive Sciences*, 8:539–546, 12 2004.
3. Aysegul Caliskan, Ahmet Cenk Andac, and Kazim Yalcin Arga. Novel molecular signatures and potential therapeutics in renal cell carcinomas: Insights from a comparative analysis of subtypes. *Genomics*, 112(5):3166–3178, 2020.
4. Hanning Chen, Ali Zakeri, Fei Wen, Hamza Errahmouni Barkam, and Mohsen Imani. Hypergraf: Hyperdimensional graph-based reasoning acceleration on fpga. *Proceedings - 2023 33rd International Conference on Field-Programmable Logic and Applications, FPL 2023*, pages 34–41, 2023.
5. Corinna Cortes and Vladimir N. Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995.
6. Abhishek Dalvi and Vasant Honavar. Hyperdimensional representation learning for node classification and link prediction. *WSDM 2025 - Proceedings of the 18th ACM International Conference on Web Search and Data Mining*, 1:88–97, 2 2025.
7. Claudia Danielmeier, Jan R. Wessel, Marco Steinhauser, and Markus Ullsperger. Modulation of the error-related negativity by response conflict. *Psychophysiology*, 46:1288–1298, 11 2009.
8. Xun Jiao Dongning Ma, Rahul Thapa. Molehd: Ultra-low-cost drug discovery using hyperdimensional computing. *Cornell University*, 2022.
9. Dominik M. Endres and Johannes E. Schindelin. A new metric for probability distributions. *IEEE Transactions on Information Theory*, 49:1858–1860, 7 2003.
10. Lydia W.S. Finley. What is cancer metabolism? *Cell*, 186:1670–1688, 4 2023.
11. E. Pusey Frady, Denis Kleyko, and Friedrich T. Sommer. Computing on functions using complex-valued hyperdimensional computing. *Frontiers in Artificial Intelligence*, 5:1–18, 3 2022.
12. Guro F. Giskeødegård, Helena Bertilsson, Kirsten M. Selnæs, Alan J. Wright, Tone F. Bathen, Trond Viset, Jostein Halgunset, Anders Angelsen, Ingrid S. Gribbestad, and May-Britt Tessem. Spermine and citrate as metabolic biomarkers for assessing prostate cancer aggressiveness. *PLoS ONE*, 8(4):e62375, 2013.
13. Ilaria Granata, Ichcha Manipur, Maurizio Giordano, Lucia Maddalena, and Mario Rosario Guarracino. Tumormet: A repository of tumor metabolic networks derived from context-specific genome-scale metabolic models. *Scientific Data 2022 9:1*, 9:607–, 10 2022.
14. Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using networkx. *Proceedings of the 7th Python in Science Conference (SciPy 2008)*, pages 11–15, 6 2008.
15. Douglas Hanahan. Hallmarks of cancer—then and now, and beyond. *Cell*, 0, 1 2026.
16. Mike Heddes, Igor Nunes, Pere Vergés, Denis Kleyko, Danny Abraham, Tony Givargis, Alexandru Nicolau, and Alexander Veidenbaum. Torchhd: An open source python library to support research on hyperdimensional computing and vector symbolic architectures. *Journal of Machine Learning Research*, 24:1–10, 6 2023.
17. Alejandro Hernandez-Cano, Namiko Matsumoto, Eric Ping, and Mohsen Imani. Onlinehd: Robust, efficient, and single-pass online learning using hyperdimensional system. *Proceedings -Design, Automation and Test in Europe, DATE*, 2021-February:56–61, 2 2021.
18. Mohsen Imani, Justin Morris, Samuel Bosch, Helen Shu, Giovanni De Micheli, and Tajana Rosing. Adapthd: Adaptive efficient training for brain-inspired hyperdimensional computing. *BioCAS 2019 - Biomedical Circuits and Systems Conference, Proceedings*, 10 2019.
19. Pentti Kanerva. Binary spatter-coding of ordered k-tuples. In *Proceedings of the International Conference on Artificial Neural Networks (ICANN '96)*, pages 870–873, Sorrento, Italy, 1996. Springer. RWCP Neuro SICS Laboratory, Swedish Institute of Computer Science.
20. Jaeyoung Kang, Behnam Khaleghi, Tajana Rosing, and Yeseong Kim. Openhd: A gpu-powered framework for hyperdimensional computing. *IEEE Transactions on Computers*, 2022.
21. Jaeyoung Kang, Minxuan Zhou, Abhinav Bhansali, Weihong Xu, Anthony Thomas, and Tajana Rosing. Relhd: A graph-based learning on fefet with hyperdimensional computing. *Proceedings - IEEE International Conference on Computer Design: VLSI in Computers and Processors*, 2022-October:553–560, 2022.
22. Jiyeon Kim and Ralph J. DeBerardinis. Mechanisms and implications of metabolic heterogeneity in cancer. *Cell Metabolism*, 30:434–446, 9 2019.
23. Denis Kleyko, Dmitri A. Rachkovskij, Evgeny Osipov, and Abbas Rahimi. A survey on hyperdimensional computing aka vector symbolic architectures, part i: Models and data transformations. *ACM Computing Surveys*, 55, 7 2023.
24. Alma Lindborg, Lea Musiolek, Dirk Ostwald, and Milena Rabovsky. Semantic surprise predicts the n400 brain potential. *Neuroimage: Reports*, 3:100161, 3 2023.
25. Haruka Namikawa-Kanai, Teruo Miyazaki, Taisuke Matsubara, Shunsuke Shigefuku, Shotaro Ono, Eiji Nakajima, Yukio Morishita, Akira Honda, Kinya Furukawa, and Norihiko Ikeda. Comparison of the amino acid profile between the nontumor and tumor regions in patients with lung cancer. *American journal of cancer research*, 10:2145–2159, 2020.
26. Annamalai Narayanan, Mahinthan Chandramohan, Rajasekar Venkatesan, Lihui Chen, Yang Liu, and Shantanu Jaiswal. graph2vec: Learning distributed representations of graphs. *arXiv preprint arXiv:1707.05005*, 7 2017.
27. Igor Nunes, Mike Heddes, Tony Givargis, Alexandru Nicolau, and Alex Veidenbaum. Graphhd: Efficient graph classification using hyperdimensional computing. 5 2022.
28. Anurag Passi, Juan D. Tibocha-Bonilla, Manish Kumar, Diego Tec-Campos, Karsten Zengler, and Cristal Zuniga. Genome-scale metabolic modeling enables in-depth understanding of big data. *Metabolites*, 12:14, 1 2022.
29. Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 10 2011.

30. Xinxin Peng, Zhongyuan Chen, Farshad Farshidfar, others, and The Cancer Genome Atlas Research Network. Molecular characterization and clinical relevance of metabolic expression subtypes in human cancers. *Cell reports*, 23:255, 4 2018.
31. Tony A. Plate. Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6:623–641, 5 1995.
32. Prathyush Poduval, Haleh Alimohamadi, Ali Zakeri, Farhad Imani, M. Hassan Najafi, Tony Givargis, and Mohsen Imani. Graphd: Graph-based hyperdimensional memorization for brain-like cognitive learning. *Frontiers in Neuroscience*, 16:757125, 2 2022.
33. Abbas Rahimi, Simone Benatti, Pentti Kanerva, Luca Benini, and Jan M. Rabaey. Hyperdimensional biosignal processing: A case study for emg-based hand gesture recognition. *2016 IEEE International Conference on Rebooting Computing, ICRC 2016 - Conference Proceedings*, 11 2016.
34. Kenny Schlegel, Florian Mirus, Peer Neubert, and Peter Protzel. Multivariate time series analysis for driving style classification using neural networks and hyperdimensional computing. *IEEE Intelligent Vehicles Symposium (IV)*, 2021. Accepted to IEEE IV 2021, Nagoya, Japan.
35. Laura Smets, Werner Van Leekwijck, Ing Jyh Tsang, and Steven Latre. Training a hyperdimensional computing classifier using a threshold on its confidence. *Neural Computation*, 35:2006–2023, 11 2023.
36. Michiel Stock, Wim Van Criekinge, Dimitri Boeckaerts, Steff Taelman, Maxime Van Haeverbeke, Pieter Dewulf, and Bernard De Baets. Hyperdimensional computing: A fast, robust, and interpretable paradigm for biological data. *PLOS Computational Biology*, 20:e1012426, 9 2024.
37. Wouter Vanhoecke, Hendrik Willems, Jan Nawrocki, and Vagenende Pieterjan. Kermit-ugent/designproject25-metabolismhdc at encoders, 2026.
38. Pere Verges, Tony Givargis, and Alexandru Nicolau. Refinehd: Accurate and efficient single-pass adaptive learning using hyperdimensional computing. *2023 IEEE International Conference on Rebooting Computing, ICRC 2023*, 2023.
39. Pere Verges, Igor Nunes, Mike Heddes, Tony Givargis, and Alexandru Nicolau. Molecular classification using hyperdimensional graph classification. 3 2024.
40. Pere Vergés, Mike Heddes, Igor Nunes, Denis Kleyko, Tony Givargis, and Alexandru Nicolau. Classification using hyperdimensional computing: a review with comparative analysis. *Artificial Intelligence Review*, 58:173–, 6 2025.
41. Zhanglu Yan, Shida Wang, Kaiwen Tang, and Weng-Fai Wong. *Efficient Hyperdimensional Computing*, page 141–155. Springer Nature Switzerland, 2023.
42. Sanggeon Yun, Nathaniel D. Bastian, Raheeb Hassan, Ryozo Masukawa, and Mohsen Imani. Missionhd: Hyperdimensional refinement of distribution-deficient reasoning graphs for video anomaly detection. *arXiv preprint arXiv:2508.14746*, 2 2026.
43. Zhuowen Zou, Kenneth Hsu, Farhad Imani, and Tajana Rosing. Hyghd: Efficient hypergraph representation learning via hyperdimensional computing. *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC)*, 6 2024.

Supplementary Information

Table 6 Mean Cross-Validated F1 Scores for Ovary Dataset: Comparison by Encoder, Model, and Dimension (D) per Threshold.

Encoder	Model	D	F1 Score per Threshold		
			Q1 / 10%	Med. / 50%	Q3 / 90%
Simple	HDC	1000	0.39	0.46	0.48
		5000	0.54	0.58	0.49
		10000	0.48	0.52	0.49
	HDC + LR	1000	0.50	0.61	0.63
		5000	0.61	0.66	0.63
		10000	0.61	0.68	0.60
	HDC + SVM	1000	0.48	0.59	0.66
		5000	0.59	0.65	0.58
		10000	0.58	0.65	0.62
Statistical	HDC	1000	0.34	0.45	0.49
		5000	0.32	0.45	0.50
		7500	0.31	0.45	0.43
		10000	0.29	0.45	0.44
	HDC+LR	1000	0.41	0.60	0.62
		5000	0.56	0.73	0.69
		7500	0.62	0.71	0.67
		10000	0.56	0.74	0.67
	HDC+SVM	1000	0.51	0.54	0.59
		5000	0.58	0.69	0.68
		7500	0.61	0.76	0.61
		10000	0.57	0.73	0.65

Table 7 Metrics of the Test Set: Simple vs. Statistical Encoder (Optimal Parameter Combinations)

Dataset	Encoder	Model	D	Threshold	F1	Acc.	MCC
Ovary	Simple	HDC	5000	Median	0.62	0.63	0.51
		HDC + LR	10000	Median	0.74	0.75	0.67
		HDC + SVM	1000	Q3	0.64	0.64	0.53
	Stat.	HDC	5000	90%	0.50	0.53	0.42
		HDC + LR	10000	50%	0.74	0.74	0.66
		HDC + SVM	7500	50%	0.76	0.76	0.69
Prostate G3-4	Simple	HDC	10000	Q3	0.62	0.61	0.21
		HDC + LR	10000	Q3	0.69	0.66	0.31
		HDC + SVM	10000	Q3	0.66	0.64	0.28
	Stat.	HDC	7500	90%	0.41	0.50	0.22
		HDC + LR	7500	90%	0.67	0.69	0.36
		HDC + SVM	7500	90%	0.69	0.70	0.39
Prostate G3-4+5	Simple	HDC	10000	Q3	0.75	0.71	0.40
		HDC + LR	5000	Q1	0.78	0.72	0.41
		HDC + SVM	5000	Q1	0.78	0.72	0.40
	Stat.	HDC	7500	90%	0.44	0.49	0.22
		HDC + LR	7500	90%	0.70	0.72	0.40
		HDC + SVM	7500	90%	0.70	0.72	0.40
Lung	Simple	HDC	10000	Q1	0.87	0.88	0.77
		HDC + LR	10000	Median	0.92	0.92	0.84
		HDC + SVM	10000	Median	0.91	0.92	0.84
	Stat.	HDC	10000	90%	0.94	0.95	0.89
		HDC + LR	10000	90%	0.95	0.95	0.91
		HDC + SVM	10000	90%	0.95	0.96	0.91
Brain	Simple	HDC	10000	Q3	0.92	0.89	0.74
		HDC + LR	10000	Median	0.97	0.95	0.87
		HDC + SVM	10000	Q1	0.94	0.91	0.77
	Stat.	HDC	10000	90%	0.89	0.91	0.79
		HDC + LR	10000	90%	0.98	0.99	0.96
		HDC + SVM	10000	90%	0.97	0.98	0.94
Kidney	Simple	HDC	10000	Q1	0.90	0.93	0.85
		HDC + LR	10000	Median	0.92	0.95	0.88
		HDC + SVM	10000	Median	0.94	0.96	0.91
	Stat.	HDC	10000	90%	0.95	0.94	0.88
		HDC + LR	10000	50%	0.95	0.94	0.88
		HDC + SVM	5000	50%	0.95	0.94	0.88

Table 8 Results for the kidney classification task using the star subgraphs encoders combined with either the AdaptHD or OnlineHD retraining method. For each combination of hyperparameters (η=initial learning rate, γ=learning rate decay) the weighted F1-score and the accuracy (Acc.) are given that were obtained after 5 fold cross-validation.

	AdaptHD			OnlineHD		
Dataset	η, γ	F1	Acc.	η, γ	F1	Acc.
Kidney	10, 0.75	0.41	0.53	10, 0.75	0.41	0.52
Kidney	10, 0.80	0.47	0.56	10, 0.80	0.42	0.51
Kidney	10, 0.85	0.57	0.60	10, 0.85	0.56	0.62
Kidney	10, 0.90	0.58	0.60	10, 0.90	0.60	0.64
Kidney	10, 0.95	0.28	0.38	10, 0.95	0.29	0.39
Kidney	20, 0.75	0.33	0.44	20, 0.75	0.41	0.51
Kidney	20, 0.80	0.40	0.49	20, 0.80	0.41	0.51
Kidney	20, 0.85	0.61	0.65	20, 0.85	0.58	0.61
Kidney	20, 0.90	0.58	0.60	20, 0.90	0.57	0.60
Kidney	20, 0.95	0.27	0.38	20, 0.95	0.32	0.41
Kidney	30, 0.75	0.34	0.45	30, 0.75	0.43	0.53
Kidney	30, 0.80	0.48	0.58	30, 0.80	0.50	0.57
Kidney	30, 0.85	0.59	0.64	30, 0.85	0.59	0.65
Kidney	30, 0.90	0.61	0.64	30, 0.90	0.59	0.62
Kidney	30, 0.95	0.33	0.42	30, 0.95	0.33	0.38
Kidney	40, 0.75	0.41	0.50	40, 0.75	0.41	0.50
Kidney	40, 0.80	0.41	0.51	40, 0.80	0.42	0.50
Kidney	40, 0.85	0.57	0.60	40, 0.85	0.59	0.61
Kidney	40, 0.90	0.58	0.61	40, 0.90	0.61	0.64
Kidney	40, 0.95	0.28	0.38	40, 0.95	0.30	0.40
Kidney	50, 0.75	0.39	0.52	50, 0.75	0.41	0.50
Kidney	50, 0.80	0.38	0.47	50, 0.80	0.41	0.51
Kidney	50, 0.85	0.52	0.58	50, 0.85	0.60	0.64
Kidney	50, 0.90	0.59	0.60	50, 0.90	0.60	0.62
Kidney	50, 0.95	0.27	0.38	50, 0.95	0.34	0.42